

1 Computer Programming (Monsoon '25) — Quiz 1

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Duration: 45 mins

1 (5 marks)

Give the bit sequence of how -56 is stored as a char (signed 8 bit) in C language.

2 (3 marks)

What will the code below do? (assuming x, y are integers that are assigned some values)

```
printf("%d", ((x ^ y) < 0));
```

3 (3 marks)

With respect to following “for” loops in C, pick (tick mark) the valid C language statement(s). Assume that there is a prior declaration of ‘i’ in all cases?

```
for (i = 0; i < 10 ; i++)
for ( ; i < 10 ; i++)
for (i = 0; ; i++)
for (i = 0; i < 10 ; )
for ( ; ; )
```

4 (3 marks)

What is the output of the following program?

```
int i = 1, j;
for ( ; ; ) {
    if (i)
        j = --i;
    if (j < 10)
        printf("CproQuiz", j++);
    else
        break;
}
return 0;
```

5 (3 marks)

What is the output of the following program?

```
int i, count;
count = 0;
for (i = 0; i < 2025; i++) {
    count++;
}
printf("%d", count);
```

6 (3 marks)

The following program rotates (cyclic shift) an array of size n by d elements to the left. You can assume that n and d are int variables defined in the scope. Fill the blanks in the code

```
int p = 1;
while (_____) {
    int last = arr[0];
    for (int i = 0; _____; i++) {
        arr[i] = arr[i + 1];
    }
    _____ = last;
    p++;
}
```

7 (3 marks)

What is the output of the following program? Note $\sim$ has higher precedence than $\ll$.

```
char x = 6;
char y =  $\sim$ x  $\ll$  2;
printf("%d\n", y);
return 0;
```

8 (3 marks)

What is the output of the following program?

```
signed char a = 100, b = 30;
signed char c = a + b;
printf("a+b = %d\n", c);
return 0;
```

9 (3 marks)

What is the output of the following program?

```
int x = 0, y = 5;
if (x++ && ++y)
    printf("Branch 1\n");
printf("x = %d, y = %d\n", x, y);
if (++x || y++)
    printf("Branch 2\n");
printf("x = %d, y = %d\n", x, y);
return 0;
```

10 (3 marks)

What is the output of the following program and what are the final values of a, b, c?

```
int a = 3, b = 7, c = 2;
if (a++ > 3) {
    if (--b < 7) {
        printf("P5: X\n");
    } else {
        printf("P5: Y\n");
    }
} else {
    if (c++ == 2) {
        printf("P5: Z\n");
    } else {
        printf("P5: W\n");
    }
}
return 0;
```

11 (3 marks)

What is the output of the following program?

```
int i, j;
for (i = 1; i <= 3; i++) {
    for (j = i; j <= 3; j++) {
        printf("%d ", i + j);
    }
    printf("\n");
}
return 0;
```

12 (3 marks)

What is the output of the following program?

```
int a[] = {4, 3, 5, 2, 1};
int i = a[0];
while (i != 1) {
    printf("%d ", i);
    i = a[i - 1];
}
return 0;
```